

MDT035003	640x480	MIPI Interface	TFT Module
Specification			
Version: 1		Date: 06/04/2023	
Revision			
1	04/04/2023	First issue	

Display Features		 RoHS compliant	
Display Size	3.5"		
Resolution	640x480		
Orientation	Landscape		
Appearance	RGB		
Logic Voltage	2.8V		
Interface	MIPI		
Brightness	300 cd/m ²		
Touchscreen	---		
Module Size	74.38 x 62.35 x 2.47 mm		
Operating Temperature	-20°C ~ +70°C	Created By	Checked By
Pinout	30 way FFC	CL	TSB
Pitch	0.50 mm	Box Quantity	Weight / Display
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* - For full design functionality, please use this specification in conjunction with the ST7703 display driver specification.(Provided Separately)

Display Accessories	
Part Number	Description
MDIB-30FP	30 way FFC/FPC 0.5mm to 30 way JST 1mm crimp header
MDIB-CC1	Interconnect board for standard pitch pinouts to fine pitch wires. Providing pinouts for 2.54 pinout. 1.27, 1, 0.845, 0.8, 0.7, 0.65, 0.62, 0.6, 0.5 & 0.3 pads.

Optional Variants	
Appearances	Voltage



1. GENERAL SPECIFICATIONS

ITEM	STANDARD VALUE	UNIT
LCD SIZE	3.5	inch
LCD TYPE	TFT/IPS/ NORMALLY BLACK/TRANSMISSIVE	
MODULE SIZE	74.38*62.35*2.47	mm
ACTIVE AREA	70.08*52.56	mm
PIXEL PITCH	0.1095*0.1095	mm
NUMBER OF PIXELS	640*480	
DRIVER IC	ST7703	
SYSTEM INTERFACE	MIPI	
VIEWING DIRECTION	ALL	O'clock
GRAY SCALE INVERSION DIRECTION	-	O'clock
BACKLIGHT TYPE	6-CHIP WHITE LED	
TOUCH PANEL	WITHOUT	

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Technical drawing of a 3.5 inch TFT LCD module, showing front and side views with dimensions and labels.

Front View Dimensions:

- LCM: 74.38 ± 0.3
- LCD: 73.08
- A.A: 70.08
- LCM: 62.35 ± 0.3
- LCD: 59.56
- A.A: 52.56
- LCM: 27.15 ± 0.5
- LCD: 16.15 ± 0.5
- A.A: 11.63 ± 0.5
- LCM: 29.5 ± 0.5
- LCD: 2 ± 0.3
- A.A: 0.3 ± 0.05
- LCM: 15.5 ± 0.1
- LCD: 14.5 ± 0.1
- A.A: 0.5 ± 0.1
- LCM: 0.3 ± 0.05
- LCD: 0.5 ± 0.1
- A.A: 0.5 ± 0.1
- LCM: 0.3 ± 0.05
- LCD: 0.5 ± 0.1
- A.A: 0.5 ± 0.1

Side View Dimensions:

- LCM: 2.47 ± 0.3
- LCD: 2.15
- A.A: 2.15
- LCM: 0.65
- LCD: 0.65
- A.A: 0.65
- LCM: 0.3 ± 0.05
- LCD: 0.3 ± 0.05
- A.A: 0.3 ± 0.05
- LCM: 0.3 ± 0.05
- LCD: 0.3 ± 0.05
- A.A: 0.3 ± 0.05
- LCM: 0.3 ± 0.05
- LCD: 0.3 ± 0.05
- A.A: 0.3 ± 0.05
- LCM: 0.3 ± 0.05
- LCD: 0.3 ± 0.05
- A.A: 0.3 ± 0.05

Labels:

- Pull tape
- 3.5" TFT
- 640*(RGB)*480
- ALL VIEWING DIRECTION
- SILICONE
- KAPTON
- CONTACT SIDE
- Stiffener PI

LED Circuit Diagram:

LED CIRCUIT DIAGRAM: 3*2 Dice

40mA @ 9V(TYP)

Brightness 300cd/m²(TYP)

Pixel Detail:

SCALE 60:1

Pixel dimensions: 0.1095 (R), 0.1095 (G), 0.365 (B)

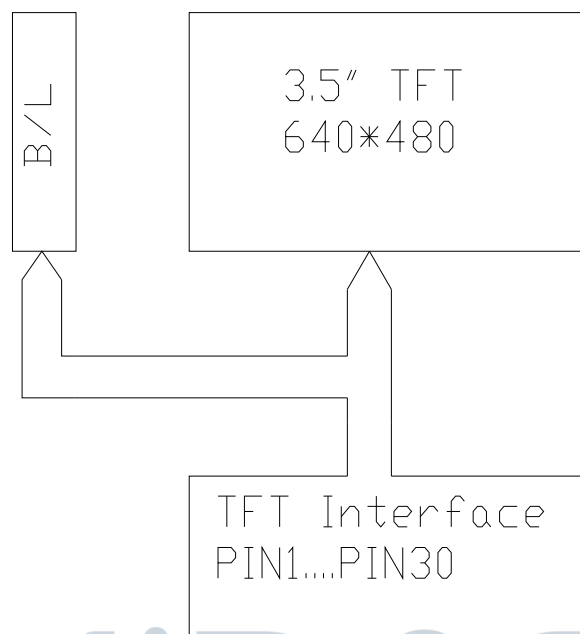
Pixel pitch: 0.1095

Pixel color: RGB

1. Unmarked tolerance is ± 0.3

2. All materials comply with RoHS

3. BLOCK DIAGRAM



MIDAS
DISPLAYS

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4. PIN ASSIGNMENT

Pin No	Symbol	Function
1	VLED+	Backlight Power Supply
2	VLED-	Backlight Power Ground
3	GND	Power Ground
4	VCI	Power supply for analog circuit
5	RESETB	Reset pin. This signal will reset the device and must be applied to properly initialize the chip.
6	NC	No connection
7	GND	Power Ground
8	IOVCC	Power supply for logic circuit
9	GND	Power Ground
10	NC	No connection
11	NC	No connection
12	GND	Power Ground
13	NC	No connection
14	NC	No connection
15	GND	Power Ground
16	DSI_CP	High speed interface clock differential signal input pins
17	DSI_CN	
18	GND	Power Ground
19	DSI_D1P	High speed interface data differential signal input pins
20	DSI_D1N	
21	GND	Power Ground
22	DSI_D0P	High speed interface data differential signal input/output pins
23	DSI_D0N	
24	GND	Power Ground
25	XR(NC)	Touch Panel Control PIN
26	YD(NC)	
27	XL(NC)	
28	YU(NC)	
29	GND	Power Ground
30	GND	Power Ground

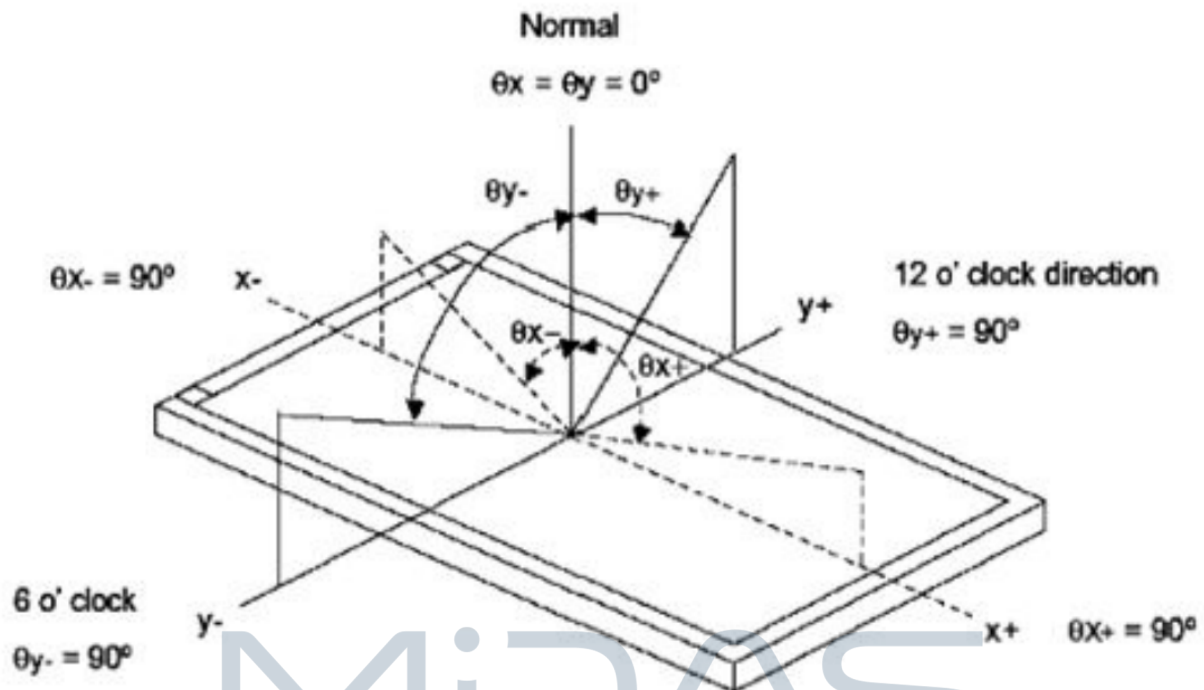


5. OPTICAL CHARACTERISTICS

Item	Symbol	Condition	Specification			Unit
			Min.	Typ.	Max.	
Luminance	L	IL =40mA	--	300	--	cd/m2
Response time	Tr+Tf	$\theta = 0^{\circ}$	-	25	50	ms
Contrast ratio	CR	$\theta = 0^{\circ}$	600	800	-	
CF Color chromaticity(CIE)	Wx	$\theta = 0^{\circ}$	-0.015	0.297	-0.015	
	Wy		-0.015	0.318	-0.015	
	Rx					
	Ry					
	Gx					
	Gy					
	Bx					
	By					
Viewing angle	Top	CR $\geq$ 10	75	85	-	deg.
	Bottom	CR $\geq$ 10	75	85	-	
	Left	CR $\geq$ 10	75	85	-	
	Right	CR $\geq$ 10	75	85	-	



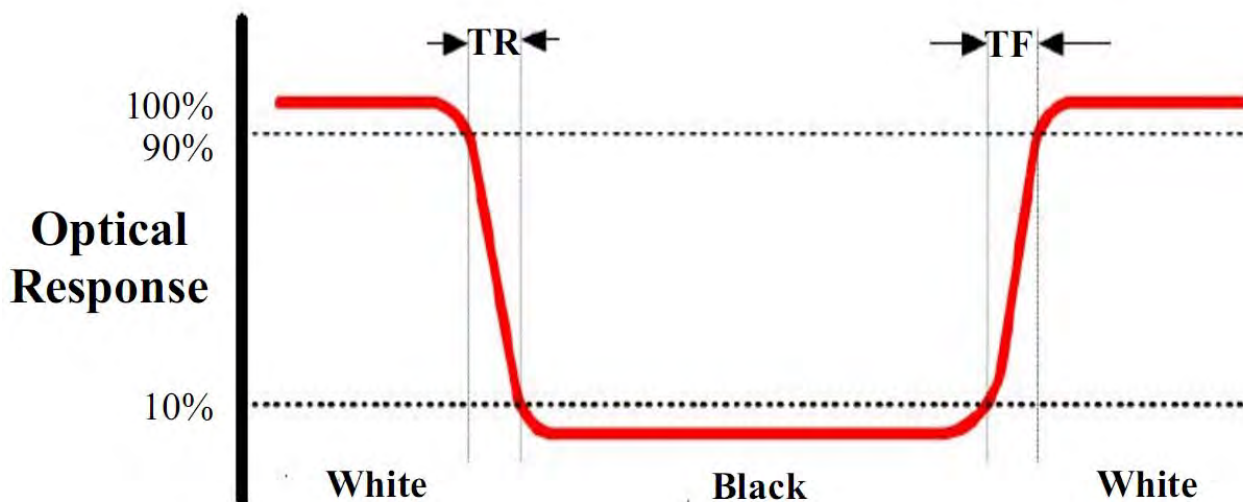
Note 1: Definition of Viewing Angle θ_x and θ_y :



Note 2: Definition of contrast ratio CR:

$$CR = \frac{\text{Luminance of white state}}{\text{Luminance of black state}}$$

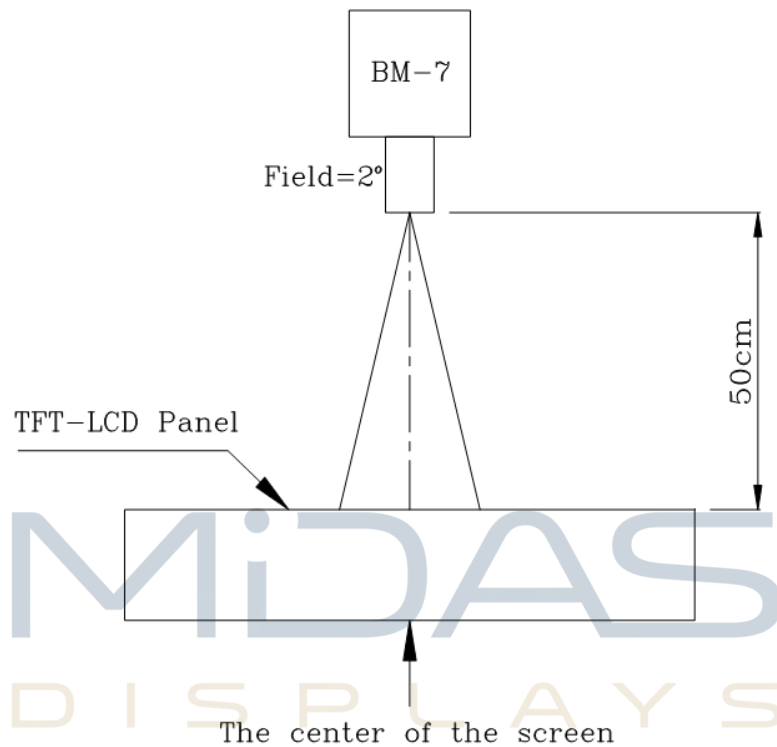
Note 3: Definition of Response Time (T_r, T_f)



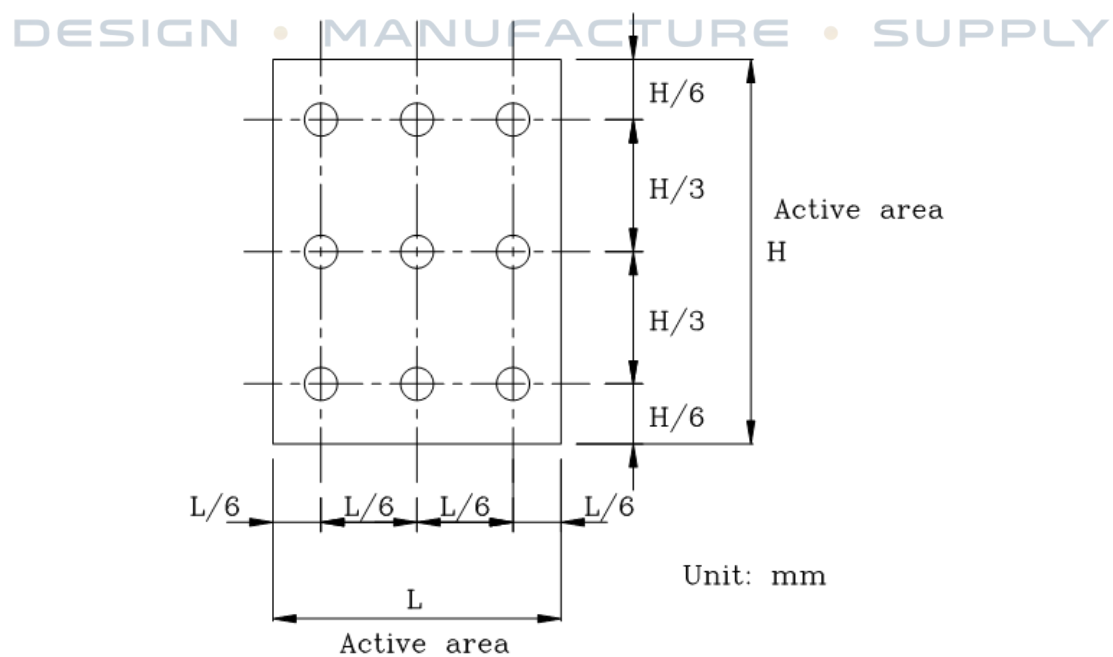
Note 4: Definition of Luminance

①The Brightness Test Equipment Setup

Field=2° (As measuring “black” image, field=2° is the best testing condition)



②The Brightness Test Point Setup



6. ABSOLUTE MAXIMUM RATINGS

Parameter	Symbol	Min.	Max.	Unit
Supply Voltage for analog	VCI	-0.3	+6.6	V
Input voltage for logic	IOVCC	-0.3	+5.5	V
Operating temperature	Top	-20	70	°C
Storage temperature	Tst	-30	80	°C

7. ELECTRICAL CHARACTERISTICS

Item	Symbol	Min.	Typ.	Max.	Unit	Applicable terminal
Supply Voltage for analog	VCI	2.5	2.8	6.2	V	VDD
Input voltage for logic	IOVCC	1.6	1.8	3.6	V	
Input voltage	VIL	VSS	-	0.3IOVCC	V	
	VIH	0.7IOVCC	-	IOVCC		
Input leakage current	ILKG	-1	-	+1	uA	

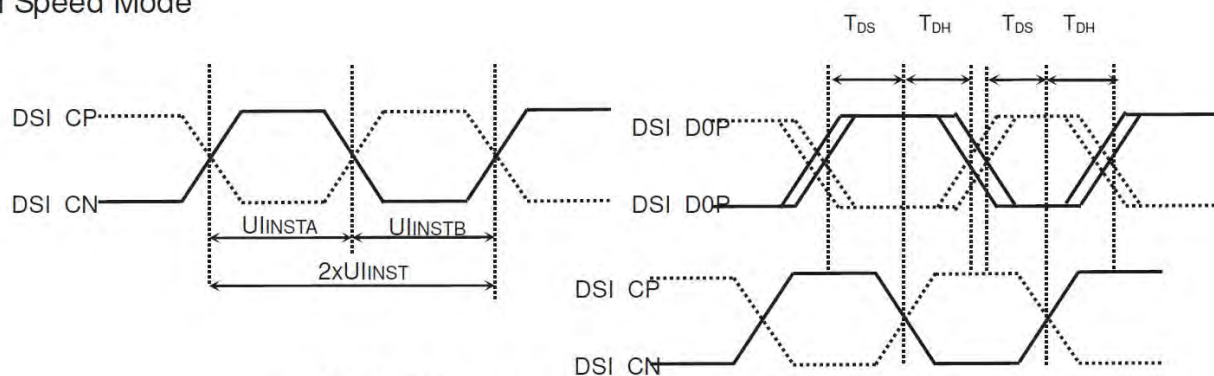
Backlight Driving conditions

Item	Symbol	values			Unit	remark
		Min.	Typ.	Max.		
Voltage for LED backlight	VF		9		V	
Current for LED Backlight	IL		40		mA	
Power consumption	P		0.36		W	
LED Life Time	-		50,000		Hr	

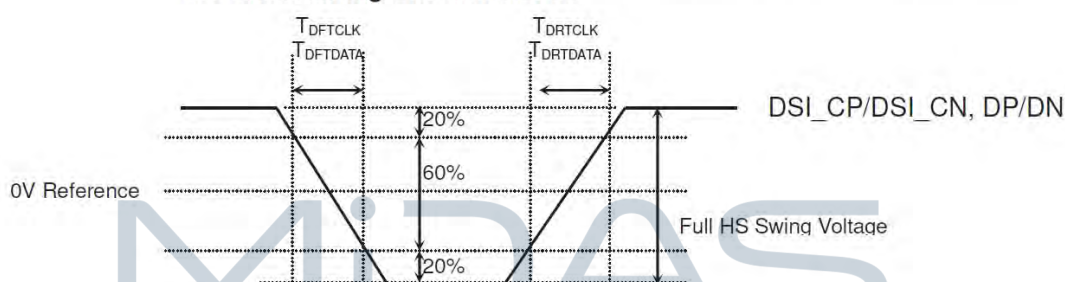


8. TIMING CHARACTERISTICS

High Speed Mode



DSI clock timing Characteristics



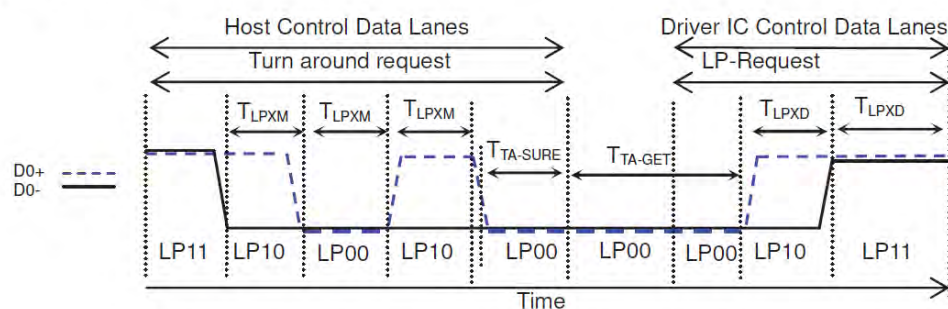
Rising and falling time on clock and data channel

(VSSA=0V, IOVCC=1.65V to 3.3V, VCI=2.5V to 3.3V, $T_A = -30$ to 70°C)

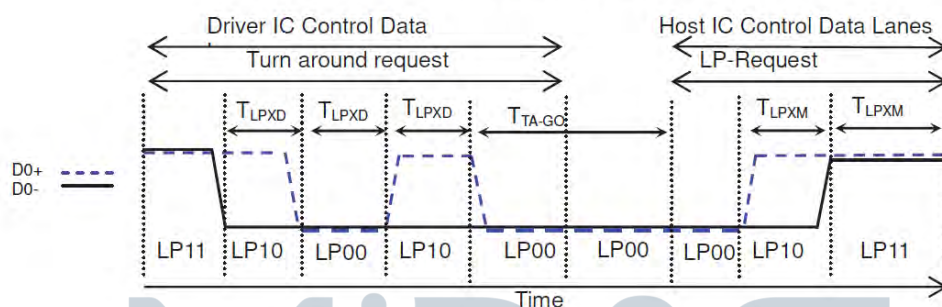
Signal	Item	Symbol	Spec.			Unit
			Min.	Typ.	Max.	
DSI_CP/ DSI_CN	Double UI instantaneous	$2xU_{INST}$	TBD	-	25	ns
	UI instantaneous	U_{INSTA} U_{INSTB}	TBD	-	12.5	ns
DP/DN	Data to clock setup time	T_{DS}	$0.15xUI$	-	-	ps
	Data to clock hold time	T_{DH}	$0.15xUI$	-	-	ps
DSI_CP/ DSI_CN	Differential rise time for clock	T_{DRTCLK}	150	-	$0.3UI$	ps
	Differential fall time for clock	T_{DFTCLK}	150	-	$0.3UI$	ps
DP/DN	Differential rise time for data	$T_{DRTDATA}$	150	-	$0.3UI$	ps
	Differential fall time for data	$T_{DFTDATA}$	150	-	$0.3UI$	ps

DSI High Speed Mode Characteristics

Low Power Mode



BTA from HOST to Display Module Timing



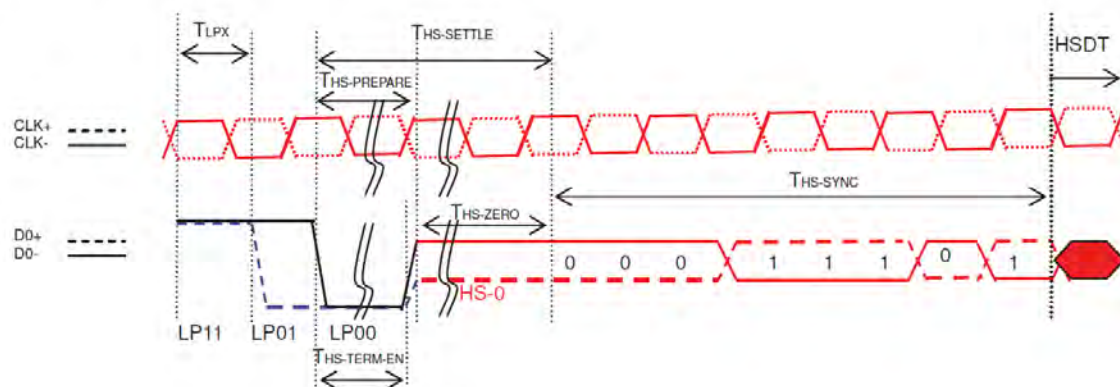
BTA from Display Module Timing to HOST

(VSSA=0V, IOVCC=1.65V to 3.3V, VCI=2.3V to 3.3V, $T_A = -30$ to 70°C)

Signal	Item	Symbol	Spec.			Unit
			Min.	Typ.	Max.	
DSI_D0P/ DSI_D0P	Length of LP-00/LP01/LP10/LP11 Host → Display module	T_{LPXM}	50	-	-	ns
	Length of LP-00/LP01/LP10/LP11 Display module → Host	T_{LPXD}	50	-	-	ns
	Time-out before the MPU start driver	$T_{TA-SURE}$	T_{LPXD}	-	$2 \times T_{LPXD}$	ns
	Time to drive LP-00 by display module	T_{TA-GET}	$5 \times T_{LPXD}$	-	-	ns
	Time to drive LP-00 after turnaround request Host	T_{TAGO}	$4 \times T_{LPXD}$	-	-	ns

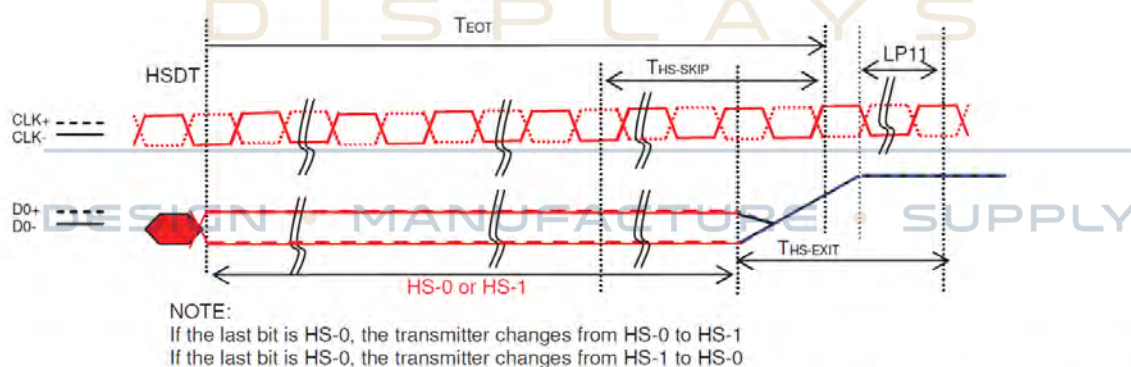
DSI Low Power Mode Characteristics

DSI BURSTS



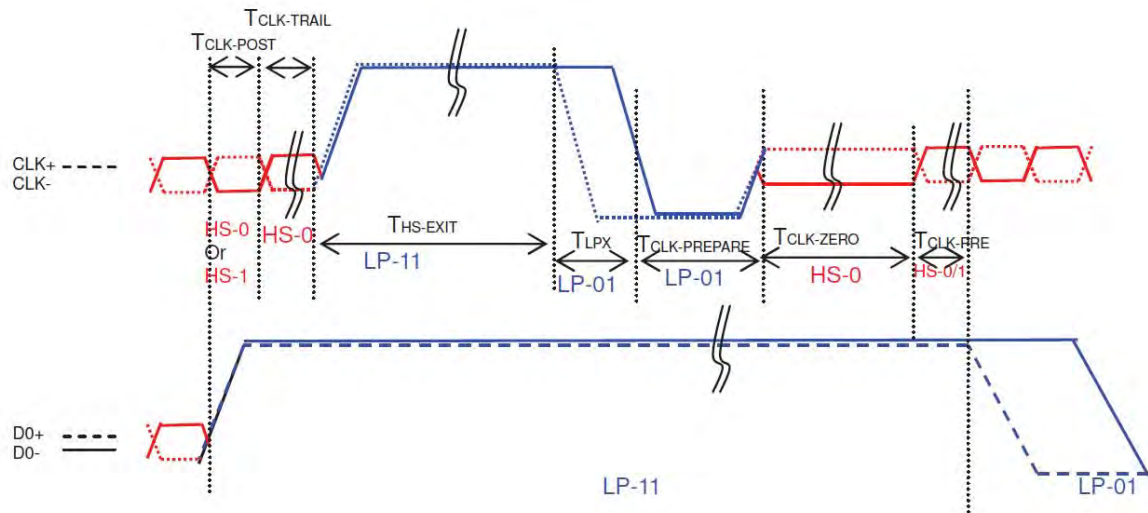
Signal	Item	Symbol	Spec.			Unit
			Min.	Typ.	Max.	
DSI_D0P/ DSI_D0P	Length of LP-00/LP01/LP10/LP11	T _{LPX}	50	-	-	ns
	Time to Driver LP-00 to prepare for HS transmission	T _{HS-PREPARE}	40+4UI	-	85+6UI	ns
	Time to enable data receiver line termination	T _{HS-TERM-EN}	-	-	35+4xUI	ns
	Time to drive LP-00 by display module	T _{TA-GET}	5xT _{LPXD}	-	-	ns
	Time to drive LP-00 after turnaround request Host	T _{TAGO}	4xT _{LPXD}	-	-	ns

DSI Low Power Mode to High Speed Mode Timing



Signal	Item	Symbol	Spec.			Unit
			Min.	Typ.	Max.	
DSI_D0P/ DSI_D0P	Time-Out at Display Module to Ignore Transition Period of EoT	T _{HS-SKIP}	40	-	55+4xUI	ns
	Time to Driver LP-11 after HS Burst	T _{HS-EXIT}	100	-	-	ns

DSI Low Power Mode to High Speed Mode Timing



Signal	Item	Symbol	Spec.			Unit
			Min.	Typ.	Max.	
DSI_CP/ DSI_CN	Time that the MCU shall continue sending HS clock after the last associated Data Lane has transitioned to LP mode	$T_{CLK-POST}$	$60+52xUI$	-	-	ns
	Time to drive HS differential state after last payload clock bit of a HS transmission burst	$T_{CLK-TRAIL}$	60	-	-	ns
	Time to drive LP-11 after HS burst	$T_{HS-EXIT}$	100	-	-	ns
	Time to drive LP-00 to prepare for HS transmission	$T_{CLK-PREPARE}$	38	-	95	ns
	Time-out at Clock Lane Display Module to enable HS Termination	$T_{CLK-TERM-EN}$	-	-	38	ns
	Minimum lead HS-0 drive period before starting Clock	$T_{CLK-PREPARE} + T_{CLK-ZERO}$	300	-	-	ns
	Time that the HS clock shall be driven prior to any associated data Lane beginning the transition from LP to HS mode	$T_{CLK-PRE}$	$8xUI$			

Clock Lanes High Speed Mode to/from Low Power Mode Timing

9. RELIABILITY TEST

NO.	TEST ITEM	CONDITIONS	
1	HIGH TEMPERATURE STORAGE	TA=+80℃	240H
2	LOW TEMPERATURE STORAGE	TA=-30℃	240H
3	HIGH TEMPERATURE OPERATION	TA=+70℃	240H
4	LOW TEMPERATURE OPERATION	TA=-20℃	240H
5	HIGH TEMPERATURE AND HIGH HUMIDITY OPERATION	+60℃, 90%RH	240H



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10. LCD MODULES HANDLING PRECAUTIONS

- The display panel is made of glass. Do not subject it to a mechanical shock by dropping it from a high place, etc.
- If the display panel is damaged and the liquid crystal substance inside it leaks out, do not get any in your mouth. If the substance come into contact with your skin or clothes promptly wash it off using soap and water.
- Do not apply excessive force to the display surface or the adjoining areas since this may cause the color tone to vary.
- The polarizer covering the display surface of the LCD module is soft and easily scratched. Handle this polarize carefully.
- To prevent destruction of the elements by static electricity, be careful to maintain an optimum work environment.
 - Be sure to ground the body when handling the LCD module.
 - Tools required for assembly, such as soldering irons, must be properly grounded.
 - To reduce the amount of static electricity generated, do not conduct assembly and other work under dry conditions.
 - The LCD module is coated with a film to protect the display surface. Exercise care when peeling off this protective film since static electricity may be generated.
- Storage precautions
When storing the LCD modules, avoid exposure to direct sunlight or to the light of fluorescent lamps. Keep the modules in bags designed to prevent static electricity charging under low temperature / normal humidity conditions (avoid high temperature / high humidity and low temperatures below 0°C). Whenever possible, the LCD modules should be stored in the same conditions in which they were shipped from our company.

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11. OTHERS

- Liquid crystals solidify at low temperature (below the storage temperature range) leading to defective orientation of liquid crystal or the generation of air bubbles (black or white). Air bubbles may also be generated if the module is subjected to a strong shock at a low temperature.
- If the LCD modules have been operating for a long time showing the same display patterns may remain on the screen as ghost images and a slight contrast irregularity may also appear. Abnormal operating status can be resumed to be normal condition by suspending use for some time. It should be noted that this phenomena does not adversely affect performance reliability.
- To minimize the performance degradation of the LCD modules resulting from caused by static electricity, etc. exercise care to avoid holding the following sections when handling the modules:
 - Exposed area of the printed circuit board
 - Terminal electrode sections.

